

of tasks simply by providing some information about how objects are typically manipulated and then showing the robot a single demo of the task. Importantly, this enables users to teach robots skills that can be automatically transferred to other robots that have different ways of moving—a key time- and cost-saving measure for companies that want a range of robots to perform similar actions.

The team tested the system on Optimus, a new two-armed robot designed for bomb disposal that they programmed to perform tasks such as opening doors, transporting objects, and extracting objects from containers. In simulations they showed that Optimus learned skills that could be seamlessly transferred to Atlas, a humanoid robot.

3D printed skin to let robots feel

Researchers at the University of Minnesota have developed a revolutionary process for 3D printing stretchable electronic sensory devices that could give robots the ability to feel their environment.



3D printing stretchable electronic sensory devices that could give robots the ability to feel their environment. (Image courtesy: <http://odishatu.in>)

“This stretchable electronic fabric we developed has many practical uses,” said Michael McAlpine, a University of Minnesota mechanical engineering associate professor and lead researcher on the study. “Putting this type of ‘bionic skin’ on surgical robots would give surgeons the ability to actually feel during minimally invasive surgeries, which would make surgery easier instead of just using cameras like they do now. These sensors could also make it easier for other robots to walk and interact with their environment.” This technology could eventually be used for health monitoring or by soldiers in the field to detect dangerous chemicals or explosives.

“While we haven’t printed on human skin yet, we were able to print on the curved surface of a model hand using our technique,” McAlpine said. “We also interfaced a printed device with the skin and were surprised that the device was so sensitive that it could detect your pulse in real time.”

McAlpine and his team made the sensing fabric with a one-of-a-kind 3D printer they built in the lab. The

multifunctional printer has four nozzles to print the various specialised inks that make up the layers of the device—a base layer of silicone, top and bottom electrodes made of a conducting ink, a coil-shaped pressure sensor, and a sacrificial layer that holds the top layer in place while it sets. The supporting sacrificial layer is later washed away in the final manufacturing process. Conventional 3D printing using liquid plastic is too hot and too rigid to use on the skin, while these flexible 3D printed sensors can stretch up to three times their original size.

IoT sensors that connect via ambient radio waves

Researchers at Disney Research’s Wireless Systems group have devised an ultra-low-power system of sensors that transmit data to a central receiver by reflecting the ambient radio waves from commercial broadcasting systems.

“Our idea is to reuse all the radio signals that are around us as a medium for transmitting data, much like sending ripples across a pond,” said Alanson Sample, associate lab director and team leader.

This approach radically reduces the power requirements of the sensor nodes because it is the generation of radio waves that consumes most of their battery power. In a demonstration in a Disney Research lab, the researchers were able to meet the tiny bit of power demand that remained by using solar cells optimised for low-light conditions.

Sample said the ultra-wide-band (UWB) approach—which backscatters all ambient sources—offers key advantages. Using multiple backscatter channels boosts the signal-to-noise ratio, substantially improving the sensitivity of the backscatter reader and decreasing dead zones. That, in turn, enables the system to operate on real-world ambient sources and substantially extends the range to 22 metres when using ambient signals from broadcast towers and 50 metres when using ambient signals generated by mobile phone up-link traffic.

“The promise of the Internet of Things (IoT) is that wireless sensors will be ubiquitous, allowing devices to sense their environments and talk to each other,” said Markus Gross, vice president at Disney Research. “As we move towards connecting the next billion wireless devices to the Internet, however, the use of batteries to power these devices will become unworkable. UWB ambient backscatter systems, which potentially could be deployed in any metropolitan area, hold great potential for solving this dilemma.”

Tiny chip extends battery life of electronics

Engineers at the University of Texas at San Antonio (UTSA) have developed a chip that can extend the battery life of low-power electronic gadgets like cellphones. The chip is about